



Prevalence of Cervical Ribs and Elongated Transverse Process Using Radiographs of Cervical Spine and Chest in Tribhuvan University Teaching Hospital

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ABSTRACT

Background: Cervical ribs and elongated transverse processes are congenital anatomical variations arising from the seventh cervical vertebra due to abnormal development of the cervical costal element. Though these anomalies are frequently asymptomatic and discovered incidentally on radiographs, they may contribute to neurovascular compression syndromes, particularly thoracic outlet syndrome, resulting in pain, paresthesia, numbness, weakness, and vascular complications of the upper limb. Despite their clinical significance, limited data are available regarding their prevalence in Nepal. The present study aimed to determine the prevalence of cervical ribs and elongated transverse processes and to evaluate the laterality (right-sided, left-sided, and bilateral) of these anatomical variations (sex wise).

Method: This retrospective, descriptive cross-sectional study was conducted in 2000 patients from February 2026 to April 2026, following ethical approval. Data were collected after evaluation of cervical spine and chest radiographs (lateral view), and were analysed. Mean value and standard deviation were used for descriptive statistics and categorical variables were presented as frequency and percentage. The level of significance was kept at $p < 0.01$. The association between presence of cervical ribs and elongated transverse process and patient's sex, were assessed using chi-square test using SPSS version 25.

Result: The prevalence of cervical ribs and elongated transverse processes were 43 (2.15%) (1.32- 2.99 at 99% CI) and 266 (13.30%) (11.34-15.26 at 99% CI) respectively. Elongated transverse processes of the cervical vertebrae are more frequently observed than cervical ribs.

Conclusion: This study demonstrates a high overall prevalence of elongated transverse processes. A significant association of prevalence of both cervical ribs and elongated transverse processes is observed with male and female.

Keywords: Cervical rib; Elongated transverse process; Prevalence.

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INTRODUCTION

Cervical rib (CR), defined as a supernumerary rib arising from the costal element of the transverse process, and elongated transverse process (ETP), representing an incomplete or rudimentary rib-like formation that does not articulate with the sternum or costal cartilage, are congenital anatomical variations of the cervical spine. These anomalies result from abnormal development of the costal elements of the cervical vertebrae, most commonly involving the seventh cervical vertebra. [1, 2] Although CR and ETP are frequently asymptomatic and are often detected incidentally during routine radiographic examinations, their presence may have considerable clinical relevance because of their potential association with neurovascular compression at the thoracic outlet. [2, 3] In particular, cervical ribs may contribute to narrowing of the thoracic outlet and compression of the brachial plexus or subclavian vessels, potentially resulting in tho-

racic outlet syndrome and related neurological or vascular symptoms. [1, 4-8] The reported prevalence of CR and ETP varies considerably among different populations and has been associated with demographic characteristics, including sex and racial or ethnic background. [4-8]. Differences in prevalence may also reflect variations in sample characteristics, imaging techniques, and diagnostic criteria used to identify these anatomical variants. The assessment of laterality is also clinically relevant, as these variations may occur unilaterally or bilaterally, with possible differences in their distribution between males and females. [1, 4-8] Despite their anatomical and potential clinical significance, there is limited published evidence regarding the prevalence and distribution of CR and ETP in the Nepalese population. Therefore, this study aimed to determine the prevalence of CR and ETP, assess their laterality, and evaluate their association with gender in the studied population.

METHODS

This descriptive and cross-sectional study was conducted to determine the prevalence of CR and ETP and to evaluate their laterality and their association with gender in the Department of Radiology and Imaging, TUTH, Kathmandu, Nepal. All the cervical spine radiographs (antero-posterior view) and chest radiographs (Postero-anterior view) were evaluated for three months period from February 2026 to April 2026 after taking ethical approval. Ethical clearance was taken from the Institutional Review Committee (IRC) of the Institute of Medicine (IOM), Tribhuvan University, (IRC reference number: 887/082/083). The data were collected from the digital records of AGFA DDR 14e detectors connected to the Hitachi X-ray machine. Patient demographic data such as age and sex were recorded from the radiology database while maintaining confidentiality. Only cervical spine radiographs (Antero-posterior view) and chest radiographs (Postero-anterior view) with adequate positioning and clear visualization of the lower cervical vertebrae (C5–C7) were selected for analysis. Radiographs with severe rotation, motion artefacts, prior cervical spine surgery, fractures, tumors, or congenital spinal deformities were excluded from the study. Presence of a cervical rib was considered when a supernumerary rib arising from the transverse process of the seventh cervical vertebra (C7), extending beyond the normal transverse process and demonstrating cortical and medullary continuity with C7 on the radiographs Figure 1.

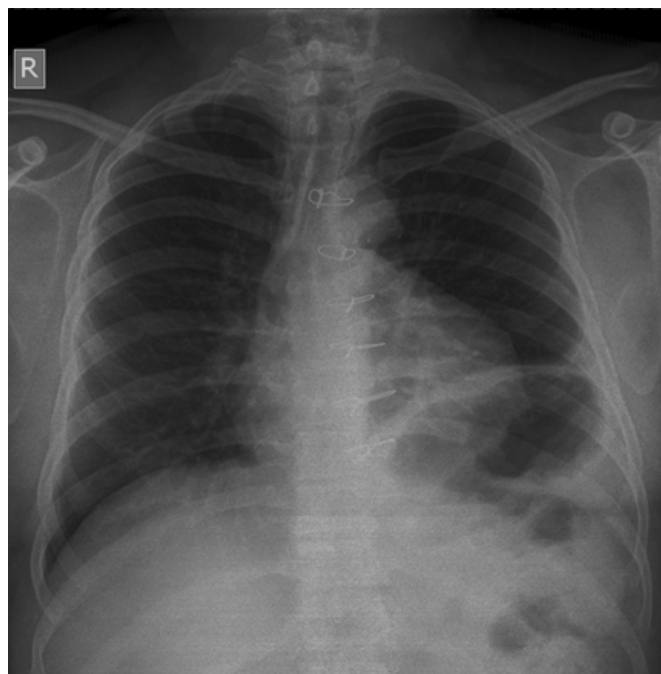


Figure 1. A chest radiograph (postero-anterior view) having bilateral cervical rib.

Similarly, presence of an elongated transverse process was considered when an abnormally long transverse process of C7 that extends laterally beyond the transverse process

of first thoracic vertebra but does not form a complete rib structure and does not show a distinct rib-like articulation Figure 2.



Figure 2. A chest radiograph (antero-posterior view) having bilateral elongated transverse process.

A total of 2000 radiographs were included in this study using total population sampling (images of all the patients meeting the inclusion criteria) during the study period. All the relevant data were collected in a self and pre-designed pro-forma and data were entered into the Microsoft Excel Worksheet and were transferred to the Statistical Package for Social Science (SPSS) program. Statistical analyses were carried out using IBM SPSS version 16 (IBM Corp., Armonk, NY, USA). Mean value and standard deviation (SD) were used for descriptive Statistics and categorical variable were presented as frequency and percentage. The level of signification was kept at $p < 0.01$ [9]. The association between presence of cervical ribs and elongated transverse process and patient's sex, were assessed using chi-square test using SPSS version 27.

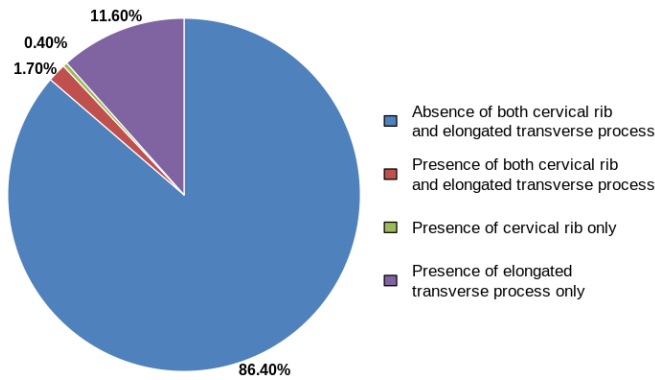
RESULTS

Among 2000 cases included in the study, 930 (46.50%) were male and 1070 (53.50%) were female. Age of the patients ranged from 5 years to 88 years, with a mean age 44 ± 17.08 years. The prevalence of both cervical rib and elongated transverse process in the same individual was 33 (1.70%) (0.96-2.45 at 99% CI). The individual with only cervical rib was 8(0.40%) and the individual with only elongated transverse process was 232(11.60%) Figure 3.

The prevalence of cervical rib was 43 (2.15%) (1.32-2.99 at 99% CI). The prevalence of cervical rib among male was 8(0.86%) (0.08-1.64 at 99% CI) while it was

Table 1. Gender and side-wise distribution of cervical rib and elongated transverse process ($n = 2000$)

Variable	Cervical rib, $n(\%)$				Elongated transverse process, $n(\%)$			
	Right	Left	Bilateral	Total	Right	Left	Bilateral	Total
Male ($n = 930$)	4 (0.43)	1 (0.11)	3 (0.32)	8 (0.86)	14 (1.50)	43 (4.62)	42 (4.52)	99 (10.64)
Female ($n = 1070$)	12 (1.12)	12 (1.12)	11 (1.03)	35 (3.27)	16 (1.49)	67 (6.26)	84 (7.85)	167 (15.61)
Total ($n = 2000$)	16 (0.80)	13 (0.65)	14 (0.70)	43 (2.15)	30 (1.50)	110 (5.50)	166 (8.30)	266 (13.30)

**Figure 3.** Distribution of cervical rib and elongated transverse process among the study population.

35 (3.27%) (1.87-4.67 at 99% CI) in female. Similarly, the prevalence of elongated transverse process was 266 (13.30%) (11.34-15.26 at 99% CI). The prevalence of elongated transverse process among male was 99 (10.64%) (8.04-13.24 at 99% CI) while it was 167 (15.61%) (12.75-18.47 at 99% CI) in female Table 1.

A significant association of the prevalence of the cervical rib was observed between male and female, (Chi-square value (χ^2) = 13.746, and p -value < 0.001). Similarly, a significant association of the prevalence of elongated transverse process between male and female was also found, (Chi-square value (χ^2) = 10.625, and p -value = 0.001) Table 2.

Table 2. Association of cervical rib and elongated transverse process between male and female ($n = 2000$)

Variable	Present	Absent	χ^2 Value	p -value
Cervical Rib (CR), $n(\%)$				
Male	8 (0.90)	922 (99.10)	13.746	< 0.001
Female	35 (3.30)	1035 (96.70)		
Elongated Transverse Process (ETP), $n(\%)$				
Male	99 (10.60)	831 (89.40)	10.625	0.001
Female	167 (15.60)	903 (84.40)		

Note. CR = Cervical Rib; ETP = Elongated Transverse Process.

Significant differences are indicated by $p < 0.05$.

DISCUSSION

Congenital variations of the cervicothoracic junction, particularly cervical rib and elongated transverse process, represent developmental anomalies resulting from

altered formation of the costal element of cervical vertebrae, most commonly the seventh cervical vertebra (C7). Although these variations are frequently asymptomatic and detected incidentally on radiographs, their identification is clinically relevant because of their association with thoracic outlet syndrome and possible compression of neurovascular structures. [1–3] The present study evaluated the prevalence of cervical rib and elongated transverse process, their laterality, and association with gender among 2000 individuals undergoing cervical spine and chest radiography in a tertiary care hospital in Nepal. In the present study, the prevalence of cervical rib was 2.15% (99% CI: 1.32–2.99), while the prevalence of elongated transverse process was 13.30% (99% CI: 11.34–15.26). Additionally, 1.70% of individuals had coexistence of both cervical rib and elongated transverse process. The prevalence of cervical rib observed in this study is comparable with previously reported studies from different populations. Brewin et al. reported a prevalence of approximately 0.74% in a London population, whereas studies from Asian populations have reported higher prevalence ranging from approximately 1–6%. [10, 11] These variations in prevalence may be attributed to differences in ethnicity, genetic background, and diagnostic criteria used for defining cervical rib. The prevalence of elongated transverse process in the present study (13.30%) was within the range reported in previous studies. Al-Kindi et al. reported a prevalence of elongated transverse process of approximately 12.3% in an Omani population, while Subhi et al. reported a higher prevalence using computed tomography-based evaluation. [1, 12] Computed tomography provides better visualization of cortical and medullary continuity and may detect subtle variations more accurately than plain radiography, which may explain differences among studies. However, plain radiography remains a widely available and practical imaging modality for population-based evaluation, particularly in resource-limited settings. The prevalence of cervical rib was significantly higher among females compared with males in the present study (3.27% vs. 0.86%, < 0.001). Similarly, elongated transverse process was significantly more prevalent among females compared with males (15.61% vs. 10.64%, $p=0.001$). Previous studies have also demonstrated a higher occurrence of cervical rib among females. Brewin et al. and Sharma et al. reported a female predominance, suggesting possible sex-related differences in developmental or genetic factors influencing cervicothoracic skeletal variation [11]. How-

ever, the exact biological mechanism responsible for this association remains unclear and requires further investigation. Regarding laterality, the present study demonstrated that bilateral cervical ribs (0.70%) were slightly more common than unilateral cervical ribs. Among unilateral cases, right-sided cervical ribs were observed more frequently than left-sided ribs (0.80% vs. 0.65%). Similar variability in laterality patterns has been reported in previous studies. The presence of bilateral cervical ribs may indicate a generalized developmental disturbance affecting both sides during vertebral segmentation. Differences in reported side predominance among studies may be related to sample size and population characteristics. The clinical importance of identifying cervical rib and elongated transverse process lies mainly in their relationship with thoracic outlet syndrome (TOS). Cervical ribs may contribute to compression of the lower brachial plexus or subclavian vessels by altering the anatomical space at the thoracic outlet. [4–8] Although most individuals with cervical ribs remain asymptomatic, recognition of these variants is important in patients presenting with upper limb neurological symptoms, vascular compromise, or unexplained thoracic outlet symptoms. Imaging plays an important role in identifying anatomical abnormalities and guiding clinical management. [13, 14] Though this study provides baseline data regarding cervical rib and elongated transverse process prevalence in the Nepalese population, the findings of this study should be interpreted considering certain limitations. First, this was a single-center study conducted in a tertiary care hospital; therefore, the findings may not fully represent the general Nepalese population. Second, evaluation was performed using plain radiographs rather than computed tomography, which may limit detection of subtle anatomical variations. Third, clinical correlation with symptoms of thoracic outlet syndrome was not performed. Future multicenter studies using advanced imaging modalities and clinical assessment may provide more comprehensive information regarding the anatomical and clinical significance of these variations.

CONCLUSIONS

This study concludes that cervical rib and elongated transverse process are relatively common congenital anatomical variations of the cervical spine in the Nepalese population. The prevalence of both variations showed significant association with gender, with higher occurrence among females. Awareness of these variations is important for radiologists and clinicians because of their potential association with thoracic outlet syndrome and their relevance during diagnostic evaluation and surgical planning.

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